

What if (Everyday) Technologies were Designed for Learning? Towards "Support for Learning" as a Design Goal for Every(day) Technology

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Abstract

What if everyday technologies were designed for productivity *and* learning? This is critical in the era of generative artificial intelligence and pervasive digital technologies, where technology design substantially shapes what and how humans are learning. This paper introduces “support for learning” as an explicit design goal for everyday technologies. To provide background, we first differentiate four relationships between learning and technology: (1) learning to use technology, (2) learning about technology, (3) learning as a primary design goal, and (4) learning through technology. Designers can intentionally facilitate the latter (a) by following established design principles for usability and user experience, but need to go beyond such known ground by (b) identifying potential learning goals, and (c) connecting to high-quality learning resources and technology designed from within the system. A future research agenda should investigate the relationships between patterns of technology design, patterns of technology use, and learning.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models**; *HCI design and evaluation methods*.

Keywords

HCI design goals, design goals, technology evaluation, human-centered design, human-centered computing, human-centered AI, humanistic design, lifelong learning

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1 Introduction

What if all digital technologies, and in particular everyday technologies, were designed with human learning in mind? Learning is a fundamental principle of human life, critical for individual well-being and for how our societies develop further. The relationship between technology and learning has received renewed significance in the era of generative artificial intelligence, where technology design substantially shapes what and how humans are learning.

Throughout the paper, we will use the example of writing and using modern technology to write to illustrate our discussion, and ask: What if you were writing a short story, preparing a paper for school, or writing a letter? You could use any digital writing tool like MS Word, LibreOffice or any $\LaTeX$ editor to do that. Further, either within these applications or separately, you could have technology check and correct your spelling, grammar and style; and you might ask modern generative AI-based tools to rephrase or elaborate on your text. Usability of these tools and their usefulness for actually writing your texts are general issues of concern in HCI. In relation to usefulness, probably the tools’ correctness in the sense of whether spelling, grammar and style suggestions are correct and improve the outcome of your writing would be checked. However, what about the effect of these tools on your capability to write – will you actually be able to spell better, understand grammar better, know better how to write a short story, prepare a paper for school, or how to write a concise letter the next time? Current evidence currently is mixed (we will discuss that later in this paper). In this paper we now ask: How could such tools be designed to better help you learn spelling, grammar, and how to write?

The paper’s discussion is more general, such that “writing” represents any everyday activity, and “writing tools” represents any technology that is used to facilitate everyday activities. By everyday activities and technologies, we mean activities that are carried out by many people in many different contexts, and technologies that facilitate them. Such technologies are widely available off-the-shelf, and by-and-large affordable by individual people.

To provide background, this paper revisits four different relationships between learning and technology that are discussed in extant HCI and adjacent literature: 1) learning to use technology, 2) learning about technology, 3) learning as the primary design goal, and 4) learning through (the use of) technology. We describe the first three relationships in Section 2. Subsequently in Section 3, we describe and develop further the fourth relationship, *learning through*

technology, that is the major concern in this paper. In Section 4, we switch perspective and ask how learning through technology could be considered at design time by setting “support for learning” as a design goal, discussing design implications and challenges for a design-oriented research agenda.

This paper makes two primary contributions: (1) a differentiation between four relationships between learning and technology, as a useful basis for talking about learning in HCI. (2) the development of “support for learning” as a design goal and evaluation dimension for every(day) technology, as an impulse for a renewed discussion around what is considered good design; and as a proposal for a novel direction for responsible technology design that aims beyond usability, efficiency, and user experience.

2 Background - Learning to use, about and with Technology

To provide a background, we start by describing the first three conceptualisations of learning in HCI: 1) learning about technology, 2) learning to use technology, 3) learning as the primary design goal.

2.1 Learning to Use Technology

This conceptualisation is about asking: How do users learn to use technology and how to design technology such that it is easily learnable?

As an *example*, consider the task of writing, and assume that a specific software product has been chosen for writing such as MS Word, LibreOffice Writer, Overleaf, etc.¹ Learning and knowing how to use a text editor concerns knowing the range of available functionalities, details of navigation menus, naming of commands, graphical icons, keyboard shortcuts, at least some details of operational mechanisms such as when and how automatic saving works, etc. Besides specifics for each software product, there are some more transferable principles, such as knowing typical icons for file opening and saving, what a “context menu” is and how to access it; that one can emphasize text by making it bold, italic or underlined is typical for most WYSIWYG editors (this is a convention, there would certainly other ways to emphasize such as making letters larger, capitalizing, etc.), or the menu ribbons in the MS Office suite.

Usability and learnability [21] are the related major concepts that *describe desirable qualities of technology*. Learnability focuses on how easy it is for a novice user to become familiar with the technology. Usability more broadly considers how easy and effective technology is to use. Learnability is understood as a sub-aspect of usability [21, 27]. *Appropriation* [22] and *instrumental genesis* [34, 44] are the related major concepts that *describe longer-term processes of how humans learn to use technology* through using it in the context of their ongoing activities. In this, there are two reciprocal processes: Humans learn how to use technology (the activity changes, and the human actor changes, e.g., their knowledge, behaviour, and attitude), and humans adapt technology (technology changes).

This relationship is seen as foundational for reasonable usage of

technology by users, as evidenced by the common practices to accompany software with help materials and documentation; and the introduction of new technology with training [9]. Further, there is a shared agreement in HCI to aim for building systems that are easy to learn and easy to use - with possibly allowing a steeper learning curve for systems geared towards experts, where efficiency after training may be more important than learnability. Design principles have been formulated both at a very generic and overarching level^{2,3} [21, Ch.7; p.366ff]; and more specific to types of interfaces such as for voice-based interfaces [57] or to types of technologies such as recently for generative AI [58].

Note that in this conceptualisation, the focus is on *learning to use the technology*, rather than learning the activity. In our example, users learn how to use MS Word, Libre Office or Overleaf rather than how to write good text. However, from the perspective of situated or socio-cultural learning theories, it is difficult to separate learning to use a tool from doing and learning the activity. We discuss this inter-relationship further in Section 3.

2.2 Learning about Technology

This conceptualisation is about asking: What are fundamental design elements and principles of interaction, what are fundamentals of computing technology, and how to teach them? We can differentiate learning about technology from learning to use technology (the relationship we discussed in Section 2.1 above) via the immediate practicality of knowledge. However, some such fundamental knowledge is certainly helpful and necessary for good technology use in the sense of as-intended, correct and responsible [21, p.51],[14].

As an *example*, consider again the task of writing. Users could learn about different interaction paradigms of editors and formats, such as text-only editors, WYSIWYG editors; and special languages such as Markdown or $\LaTeX$; principles of how such editors and languages work in relationship to layouting and across platforms; different file-types and their relationship to conceptual content-types such as text, image, audio, video; file versioning; how storage works on a computer etc. Some of this knowledge is actually useful when using digital tools for writing, but a lot may not be.

A user’s *mental model* of technology is the major concept which describes how a user understands how a particular system works [14, 21], i.e. by mental model a cognitive structure internal to a user is meant. Mental models always exist. They are built based on a user’s practical experience with systems perceived as similar, based on the experience with this particular system, and based on a user’s general knowledge, which may have been acquired in formal education or in other social environments. Naturally, mental models can be arbitrarily wrong. Further, *digital, computational and AI literacy* [37, 55] are the names under which HCI and adjacent communities discuss and investigate what are relevant knowledge and competencies regarding computing technology, and how best to teach it.

Despite the wide acknowledgement that most technologies do require some fundamental literacy for reasonable usage, learning about technology is not a key concern within HCI. Nonetheless, there is some HCI research that uses technology (the focus of the

¹Knowing the available choices is typically not considered under the header of knowing how to use technology, but under the header of knowing about technology, and we will discuss this perspective in the next subsection.

²D. Norman: <https://www.nngroup.com/articles/ten-usability-heuristics>

³B. Shneiderman: <https://www.cs.umd.edu/users/ben/goldenrules.html>

next subsection) in order to teach about technology (the focus of this subsection), such as Virtual Reality environments to teach computer science concepts [43]; or the conceptualisation that explainability functionality can be used to teach fundamentals of automated reasoning or recommendation [37].

2.3 Learning as the Primary Design Goal

This conceptualisation is about educational technology, and asking: How to design for learning as the primary function of technology, and how do people use such (educational) technology? This question is shared by the research communities in educational technology, educational and learning sciences, and HCI, albeit with slightly different foci.

As an *example*, let us ask how technology can be designed to facilitate learning to write? There are of course online courses e.g., for creative writing, story-telling, or academic writing. Here, one could distinguish between the underlying technological frameworks of online courses, which have been designed to support teaching, learning, and accreditation in general and not only for learning how to write; and the single courses in which the content has been designed with the affordances of these platforms in mind. A bit further out in terms of technology that is off-the-shelf-available, intelligent tutors could actively teach and feedback writing [5], conversational agents could moderate collaborative writing [31], and AI-based writing feedback could be used in traditional educational settings with a teacher as a complement [32].

In general, technology can play many different roles in learning and education⁴: Technology could be used to document learning activities and results, could support the distribution and consumption of digital learning materials (including Learning Management Systems - LMSs and Massively Open Online Courses - MOOCs), could support communication between learners, teachers and educational institutions [53], provide virtual simulations in order to support experimentation that is safer, cheaper or impossible in reality [18], provide data analytics to derive insights about learning activities [4, 49], recommend learning materials and people [39], could guide learning activities maybe aiming to mimic aspects of good human teachers [1, 4], or could aim to provide playful learning environments (e.g., [16]).

Interactive systems that are designed primarily for learning still need to be usable and learnable, and users will need some fundamental digital literacy, or even data and AI literacy in the case of learning analytics or AI-based systems. As technologies designed for a specific purpose, however, educational technologies need to consider specifics of how humans learn. While there is not the same tradition in educational technology literature to synthesize design principles, we still make a rough attempt at synthesis by pointing out three very overarching sets of principles: 1) Organise and multimodally represent learning content in a way that is conducive to learning, following e.g., principles of multimedia learning theory [40]. 2) Design the digital environment in a way conducive to learning, e.g., following established HCI design principles for achieving usability and user experience, and extending with learning-specific

design considerations such as emotional design [56]. 3) Adapt learning content, tasks and representation based on users' knowledge, behaviour (e.g., strategies, mistakes made), and affect; provide feedback, scaffolding, encouragement, and guide-self-regulation [1] - in short: imitating aspects of what good teachers do, and supporting good teachers in realising this ideal behaviour.

Note that this conceptualisation distinguishes itself from others in that it does not make a particular statement about what is learned (usage of technology, fundamentals about technology), nor about the way in which learning (e.g., formal learning vs. informal learning) should happen.

3 Learning through Technology

This conceptualisation deals with technology mediating learning even when learning hasn't been an explicit design goal for the used technology, simply due to being used as a tool in human activity. Investigating, exploring and further developing this conceptualisation is crucial, because everyday technologies shape users' learning, intentionally or not, due to digital pervasiveness and automation. Arising questions are: How do technologies (re-)structure human activity? How do people use (everyday) technology intentionally for learning? What conceptualisations about the activity are implicitly learned due to characteristics of technology design? What can designers of everyday technologies do to engage users in learning, and facilitate and support learning about the activity and becoming an expert in it, even if such learning and development of expertise is not the primary design goal?

As this relationship is the focus of the present paper, we extend the discussion of an example with a gap identification and examples for possible solutions, before laying out the learning-oriented conceptual groundwork in this section. The design-oriented perspective is then placed in the next Section 4.

As an *example*, let us go back to writing. One category of technologies *through* which people learn to write are the writing tools themselves - by using MS Word, LibreOffice, $\LaTeX$ and a suitable $\LaTeX$ editor such as Overleaf etc. to write, people learn how to write. Secondly, there are many functionalities or stand-alone tools that support writing by taking over (parts of) the task of writing, such as spelling, grammar and language style checkers, translation technologies and AI-based writing assistants [33]. To date, evidence is not clear on whether such functionalities rather support learning by modelling good spelling, grammar or writing, or whether they hinder learning by automating precisely those tasks that should be learned ([33] in general, [36] for spelling and grammar checking). However, clearly they *mediate* learning in the sense of shaping the learning that happens: they make a difference. Thirdly, there is a plethora of digital tools that allow accessing information and knowledge resources, such as browsers, Internet search engines, social media, or any communication technology to ask peers, teachers, or experts questions about writing, to give just a few examples.

What is missing? The learning described above happens *through* technology in the sense of technology mediating and shaping the activity in which learning occurs, and thereby the learning processes [34]. However, the design of the used tools is not geared towards learning: Spelling, grammar and style-checkers have been designed to support the production of texts with less errors - and

⁴<https://viktoriamammer.wordpress.com/2019/01/29/what-can-computer-technologies-do-for-learning-the-roles-and-functions-of-technologies-in-learning/>

this they do [50] - but they do not support users in learning how to produce texts with less errors (ibid). Autocomplete features have been designed to speed up writing, but makes text less original (“more predictive” [3]). Similarly, generative AI-based tools have been designed for (experimenting with) re-writing [46], but not to support users in learning how to re-write or to critique existing texts as the first step towards re-writing; and of course general information access or communication technology has primarily been designed to give access and to connect. There are missed design opportunities here: Writing tools could incorporate functionality that opens up pathways to learning: a “learning mode” for spelling, grammar and style could mark mistakes but not immediately offer the solution; an integration to flashcard tools could allow writers as learners to take note of words they often mis-spell, given the heterogeneous quality of content on the web, style checks and genAI-based rephrasing could link to a curated selection of educational resources on writing ranging from layouting to style recommendations for different genres, and proactive reflective hints or specific feedback to text elements could be given to the text written - to give just a few first examples.

Sociocultural learning theories such as activity theory [28, 34] understand learning as happening inseparably intertwined with human activity. A central insight is, that an activity impacts both the subject (the human, who intentionally carries out an activity) and its object (that towards which human activity is directed). In our example of writing, this means that when a writer writes a text, the writer is the subject and the text is the object of the activity. Through the activity of writing, both the object (=the text) and the subject (=the writer) are changed. We have already introduced instrumental genesis above (Section 2), and have there highlighted it as describing the socio-cognitive process that accompanies technology use and constitutes both of adapting technology use (the activity changes, and subsequently the human actor changes) and technology (technology changes). As instrumental genesis builds on activity theory, there is the implicit understanding that through using new tools in an activity, users learn to work with new tools (humans learn about technology), thereby re-arrange their work processes (the activity changes) and subsequently also re-arrange their ways of thinking about the task (humans learn about the activity). Referring back to the spectrum of learning along dimensions proposed by Eraut [23], the learning conceptualised here is mostly unconscious, unplanned and implicit in the users’ activity. Activity theory has already been extensively positioned in relationship to design [34]. Further theories that conceptualise learning in relationship to ongoing human activity - albeit with different foci and theoretical underpinnings - are experiential learning [35], workplace learning and informal learning [23, 28], and the development of expertise (see e.g., [15]).

From such a perspective, learning happens because technology structures and shapes human activity, and humans learn through activity. Nonetheless, many kinds of technologies that are now widely used in everyday activities are problematic for learning: tools that facilitate access to resources have the issue that they depend on the quality of the content or knowledge of the person they are connecting the learner with. Tools that automate part of the activity risk to disengage users and foster uncritical delegation of tasks [24, 52]; and particularly generative AI-based tools may be perceived as

being human-like [48] and therefore model how an activity should be done (observed in [6] in the context of a creativity task). This carries long-term risks of deskilling users, and when generative AI-based tools show unwanted or biased behaviour. Beyond this, there are concerns that web-scale platforms are fundamentally designed for and conducive to shallow cognitive engagement [13] and for users’ continuously seeking new excitement in the form of new (micro)content and immediate feedback [2], which in turn harms attention and deep engagement. Naturally, users have agency, e.g., in deciding how to use technology and configuring their activities, but this does not absolve technology designers of their responsibility.

4 Towards Setting “Support for Learning” as a Design Goal

Design goals highlight what is considered important about technology: what do designers aim to achieve when designing technology, and how will it be assessed? Over time, HCI has expanded its design goals from basic usability to broader considerations such as usefulness, user experience, and socio-technical integration [7, 10, 11, 30]. Just as socio-technical integration and user experience emerged as essential aspects of interaction design that complement concerns of usability and efficiency of interfaces, we propose to continue this trajectory of increasingly sophisticated design goals by setting “support for learning” as an explicit design goal.

Others before us have argued that due to the increasing permeation of our lives with digital technology, our relationship with technology changes [25]; and hence new ambitions in designing technology are necessary [45]. Example concepts and research agendas are positive computing [12], slow technology [29], or reflective informatics [8], each of which conceptualises and motivates slightly differently the need to set design goals beyond the utilitarian, such as psycho-social well-being, mental rest and reflection. Rogers [47] has prominently argued that technology should be “designed not to do things for people but to engage them more actively in what they currently do” (ibid, p. 406). Even more recently and in a theoretically well-grounded concept development, Ren et al. [45] have called on HCI to take more responsibility for human growth and becoming [45], and have to this purpose developed a conceptual framework for human-engaged computing. This has two aspects relevant for the present discussion. The first is, to design for engaged humans, who are capable and willing to learn, be creative, solve (complex) problems; and are attentive to and satisfied by the task they are carrying out. This concept of an engaged user is very aspirational and inspirational in terms of expectations towards humans as technology users, taking the stance that there is a “fundamental human capacity which is the neglected potential to engage out of mere necessity and to find satisfaction in the state of unmediated task engagement itself [...] when attention to the task is not divided by [...] distractions” (ibid, p. 53). The reciprocal aspect is that designers could and should take responsibility for *supporting users in becoming engaged humans*.

In this sense, the framework of human-engaged computing is an important design-oriented ground on which the present work builds. Building on such groundwork, we can now ask: *What technology characteristics, design patterns and principles in everyday technology facilitate rather than hinder learning?*

4.1 Design Principles for “Support for Learning”

From the observation that we aim to design for learning through technology in the sense of learning in the context of activity, we can draw a first fundamental design principle: Technology needs to be designed well in reference to this main activity, i.e. needs to support users’ task completion and productivity. Thus, as the basis for designing for learning, design teams need to *(a) ensure high-quality of design by following established design principles of usability and user experience*. This includes considering of socio-technical integration of technology, and aesthetics. There are also learning-directed arguments that support this deduction: many usability guidelines are based on the notion of reducing extraneous cognitive load, i.e. that cognitive load that is not related to the task at hand [54]; and also for the goal of human-engaged computing, Ren et al. [45] set “[to] draw humans into focused attention” (ibid, p. 56) as a design principles. This will be directly conducive to learning. Finally, a positive user experience contributes to a positive affective state of the user, which is also considered to be conducive to learning [56]. However, it is possible to go beyond such known ground. In order to identify further design principles from extant literature, we look at the main typical differences between “learning through technology” and learning in educational settings. These are the presence of a curriculum and set learning goals as giving direction to learning, and the presence of curated learning materials and teachers, for giving direction, explaining the domain, and giving feedback. From this we draw two more design principles: Design teams should *(b) identify potential learning goals* that users might be able to or wish to follow alongside usage of the tool. Setting explicit learning goals is considered best practice in instructional design as well as in self-directed learning [19, 59]. Goals clarify what is important, and direct focus. While this may be easier for more task-specific tools, also for general-purpose tools this seems possible. In the example of writing and writing tools used throughout this paper, such learning goals could be for instance to know and be able to apply correct spelling and grammar; or to know and be able to implement genre-appropriate language and structure. Finally, designers should *(c) connect to high-quality learning resources including AI-based learning technology from within the system* which target the identified learning goals. By resources, we here conceive of the full spectrum of learning materials, from static and printed books which can be referenced in the traditional manner, to web-based, interactive multimedia content, to proactive, adaptive and AI-based educational technology, to knowledgeable people or institutions that might provide the interested user with more guidance. This design principle is therefore about connecting users as learners to content and to guidance - be that in the form of knowledgeable persons, or technology. The level of authoring done by the design team could range from providing a curated list to existing resources and systems, to newly developing resources; and “connecting” could mean just providing pointers to a curated list of resources, or embedding proactive, adaptive AI-based tutoring or guidance functionality within the system. Such curation in the sense of providing high-quality learning resources, or links to them, is important in an era of massively available online resources of heterogeneous quality.

4.2 Challenges and Directions for Future Research

What should a future research agenda focus on, which takes seriously the responsibility to consider human learning in technology design in order to contribute to the vision for technology to augment human agency and capabilities?

The major need for knowledge that we see is to *understand better how different design patterns of technology relate*, via making some ways of using technology in activity easier and more obvious than others, *to patterns of technology use, and how both relate to human learning*. Such research must be design-oriented and socio-technical, focusing on patterns of technology design and usage. The focus on patterns is crucial: it is both more fine-granular and more generalizable than a lot of extant research. Thus, it must go beyond general statements about the “usage of ChatGPT” or “usage of social media” and aim to identify specific patterns in the investigated technology and use.

Such research must also investigate potential trade-offs between facilitating productivity and learning. However this trade-off is not straightforward, as users may (have to or want to) prioritise productivity and learning depending on a multitude of factors, and do this in a multitude of ways. Overall, technology should facilitate human agency and capabilities both in the “now” (productivity) and in the future (learning). There is reason to be optimistic that design can achieve this. To give three examples: 1) there is renewed interest in investigating the balance between automation and user control in design in the context of generative artificial intelligence (general references: [51, 58], creativity: [17, 38], writing [46]). Prioritising differently productivity or learning could lead to choosing different system behaviour at different levels of automation vs. user control. 2) Design patterns geared towards user control and experimentation within the activity could provide an anchoring point for designing for reflection as a learning strategy embedded within and connected to ongoing activity [26, 42]. 3) Vice versa, some knowledge-intensive activities could benefit from educational technology interventions to facilitate productivity. For example, Disch et al. [20] have connected educational technology design patterns with standard features of open science platforms to lay out the space of design opportunities for supporting reading in open innovation scenarios. More such systematic, integrative, and synthetical work across HCI, the learning sciences, psychology and adjacent disciplines is needed.

Such building of a knowledge base must be accompanied by conceptual and methodological groundwork that deepens both the learning-theoretical and design-theoretical foundation, and builds evaluation approaches and metrics: In the same manner as including usability and user experience as measurements is by now standard (at least in a very simple manner), measuring learning must become standard in user studies. For example, a simple first approach in experimental between-participant designs may be to follow the experimental task with a post-task in which both the experimental and control group do a similar task without tool support, as shown by Bangerl et al. [6] in the context of a creativity task. Finally, designing for productivity and learning will need to be critically reviewed, as there may be unintended consequences in

practice when mixing purpose, as e.g., discussed around data-driven technologies for professional learning [41].

5 Conclusion

This paper develops and presents “Support for Learning” as a fundamental and intentional design goal in everyday technologies. By differentiating four relationships between learning and technology – learning to use technology, learning about technology, learning as a primary design goal, and learning through technology – we have highlighted an often-overlooked opportunity: designing technology to actively facilitate ongoing, situated, and implicit learning during everyday activity. We proposed that embedding learning as a design goal requires designers to: (a) prioritize usability, usefulness, and high-quality user experience as the baseline; and then going beyond such known ground by (b) identifying potential learning goals at design time; and (c) connecting to high-quality learning resources and technology from within the system. While these principles offer a pathway to enhancing human capabilities through interaction, they also raise challenges regarding how learning-supportive design can be systematically implemented and evaluated. Looking ahead, further research is needed to investigate how specific design patterns influence learning, how to balance automation and human skill development, and evaluation metrics and approaches for studying learning in general technology user studies.

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